A

Drifting peak (same physical D_s)

Jumping peak (same rate)

Jumping peak (rate ∝ perimeter)

Time (×10³)

C20

C60

Heading $C(\ell)$

$\alpha = 2$

C = 20

C = 60

$\alpha = 0$ (curves coincide)

$\alpha = -1$

Lag (×10³)

B

C

Drift only

Drift + same-rate jumps

Drift + perimeter-rate jumps

drift

same-rate jumps

perimeter-rate jumps

Local exponent α

Input fraction f

Drift

+ same-rate jumps

+ perimeter-rate jumps

$f_{1/2}/f_{1/2}(C_{20})$

$\propto C$

Circumference C

C

Wave-pinning

Holmes

Spring

Goryachev

C; Spring ×20 in fold panel

D

Wave-pinning, $f = 0.03$

Holmes, $f = 0.02$

Spring, $f = 0.005$

Goryachev, $f = 0.01$

switch

$\alpha_{20,60} = 2.1$

95% CI [1.9, 2.3]

$\alpha_{20,60} = 1.4$

95% CI [1.3, 1.5]

$\alpha_{20,60} = 1.8$

95% CI [1.7, 1.9]

$\alpha_{20,60}$ unavailable

E

Unforced

Forced

D_s (physical)

λ_{eff}

$k_{\text{pred}}/k_{\text{obs}}$

Not estimable

Multiple-front fraction

Figure 4. Null models for size-dependent heading persistence

A. Simulated drifting and relocating peaks on small and large rings. Grayscale kymograph widths follow circumference and time increases upward; each pair shares its declared seed. Ensemble heading autocorrelations below use small-ring solid and large-ring dotted curves. Fixed physical drift, same-rate relocation and perimeter-rate relocation give size exponents 2, 0 and −1, respectively.

B. Simulated mixtures illustrate how relocation erodes a geometric size advantage. Input fraction is f; the illustrative dependences are $D_s = 8.8f^2$ and $\lambda = 135f^4$, with perimeter-rate relocation multiplying λ by C/40. The decay rate is $k(\ell) = -\ln C(\ell)/\ell$; α is minus the local slope of ln k against ln circumference. Noise tolerance $f^{1/2}$ is the first downward crossing of correlation 0.5 at the fixed reference lag. Gray shades encode increasing circumference. Line styles distinguish mixtures in the tolerance-fold panel; its dashed guide is proportional to circumference.

C. Local exponents use model hues and light-to-dark circumference ramps. Markers show designed inputs, open markers identify forced conditions, vertical intervals show pointwise 95% uncertainty, and outward triangles mark off-scale values or tails. The right panel shows tolerance folds with model-specific markers; Spring circumference is multiplied by 20. 125/260 neighborhoods resolve; Goryachev contributes 11. Largest-size tolerance folds are Wave-pinning 2.373, Holmes 2.155, Spring 3.088, Goryachev 4.122.

D. Matched first-replicate model histories use the selected designed inputs, with physical widths and shared grayscale activity scales within each model. Black outside ticks identify persistent dominant-region replacements. The ensemble curves below include end-to-end exponents and 95% intervals where estimable. At every selected input, small-ring reference-lag correlation is below 0.4 and large-ring correlation exceeds 0.6; the enlarged rings retain heading longer.

E. Columns distinguish models; rows show physical drift diffusivity, effective replacement rate, predicted/observed decay-rate ratio and multiple-front fraction. Circumference uses the same ramps; open markers identify forced conditions. The prediction is $(2\pi/C)^2 D_s + \lambda(1 - \text{mean} \cos \Delta\theta)$, with unity as the ratio reference. Predicted rates lie within 15% of observed rates in 42 of the 42 estimable conditions. Goryachev has no estimable prediction/observation ratio. Undefined values and empirical zeros on logarithmic axes are omitted with reasons recorded. Multiple-front fractions remain descriptive. Time and length retain each model’s native units throughout.